| Título | Abstract | Link | Referência | Local de Publicação | Método de busca | RQs | OBS | Considerações |
|---|---|---|---|---|---|---|---|---|
| LLMs as Design Partners for AI-Based System Patterns: An Empirical Evaluation | The growing integration of Artificial Intelligence (AI) components into software systems challenges engineers to design architectures that combine learning, reasoning, and reliability. This paper presents an empirical evaluation of how Large Language Models(LLMs) can assist in designing AI-based systems through pattern generation. We analyzed 65 documented design patterns using three state-of-the-art LLMs—Gemini 1.5 Flash, Claude 3.5 Sonnet, and o1-mini—applying a mixed-method framework that combines quantitative similarity analysis, expert qualitative judgment, and blind human evaluation. Results show that LLMs generated solutions equivalent or superior to literature-based ones for most patterns,revealing distinct model strengths: o1-mini achieved the highest alignment with existing designs, while Claude 3.5 Sonnet received the strongest human preference. The study contributes a replicable methodology for assessing LLM reasoning in software design and highlights the potential of LLMs as co-creators that complement human creativity in the engineering of AI-based systems. | LLMs as Design Partners for AI-Based System Patterns: An Empirical Evaluation | Proceedings of the IEEE/ACM 5th International Conference on AI Engineering - Software Engineering for AI | Felipe Rodrigues de Oliveira, Felipe Vasconcelos De Souza, Ana Luiza Bessa De Paula Barros, and Paulo Henrique Maia. 2026. LLMs as Design Partners for AI-Based System Patterns: An Empirical Evaluation. In Proceedings of the IEEE/ACM 5th International Conference on AI Engineering - Software Engineering for AI (CAIN '26). Association for Computing Machinery, New York, NY, USA, 12–22. https://doi.org/10.1145/3793653.3793760 | CAIN '26 | Artigo do Felipe | **RQ:** To what extent can LLMs discover or generate design pattern solutions for AI-based systems as effectively as humans? | Artigo do Felipe | |
| Large Language Models as Assistants in Software Architecture Design | To achieve desired software quality attributes, architects must evaluate and choose from a myriad of design concepts, patterns, and technologies. This article explores how large language models can assist in this decision-making process. | Large Language Models as Assistants in Software Architecture Design | IEEE Journals & Magazine | IEEE Xplore | H. Cervantes, Y. Cai and R. Kazman, "Large Language Models as Assistants in Software Architecture Design," in IEEE Software, vol. 43, no. 4, pp. 87-95, July-Aug. 2026, doi: 10.1109/MS.2026.3663353. keywords: {Artificial intelligence;Software architecture; Education;Chatbots;Systematic literature review;Security;Scalability;QR codes; Decision making;Software quality}, | IEEE Transactions on Software Engineering | Busca pelos artigos do Rick Kazman (autor referência em arquitetura de software) | **RQ1:** Can an LLM identify appropriate design concept alternatives for a given design problem?<br>**RQ2:** Is there a consensus between different LLMs on the selection of an alternative for a design problem?<br>**RQ3:** What is the relationship between the design decisions made by humans and LLMs given the same problem description? | | - Artigo com uma proposta similar (no contexto de decisão de arquiteturas por LLMs em problemas específicos)<br>- Pelas RQs e abstract creio que seja um artigo para se levar em consideração |
| Do Large Language Models Contain Software Architectural Knowledge? An Exploratory Case Study with GPT | Architectural knowledge (AK) of existing systems is essential for software engineers to make design decisions. Recently, Large Language Models (LLMs) trained on large-scale datasets, including software repositories, have shown promise in embedding knowledge and answering questions. However, LLMs have not been evaluated for their abilities to answer questions about AK, leaving doubts about their accuracy. This paper assesses GPT, a leading LLM, by evaluating its responses' accuracy, quality, and trustworthiness on the AK of the large scale open-source system HDFS. We conducted an exploratory case study with 14 software engineers who posed questions to GPT and compared its responses to a predefined ground truth. The engineers rated GPT's answers with moderate quality and trustworthiness. Our findings on GPT´s accuracy indicates moderate recall but lower precision, especially in identifying quality attribute solutions and design rationales. These results suggest that while GPT and similar models can provide initial insights into AK, expert validation remains necessary for reliability. This study underscores LLMs' potential and limitations to discover software AK. | Do Large Language Models Contain Software Architectural Knowledge? : An Exploratory Case Study with GPT | IEEE Conference Publication | IEEE Xplore | M. Soliman and J. Keim, "Do Large Language Models Contain Software Architectural Knowledge? : An Exploratory Case Study with GPT," 2025 IEEE 22nd International Conference on Software Architecture (ICSA), Odense, Denmark, 2025, pp. 13-24, doi: 10.1109/ICSA65012.2025.00012. keywords: {Knowledge engineering;Accuracy; Software architecture;Large language models;Computer architecture;Software; Software reliability;Architecture knowledge;Architecture design decisions;Large Language Models; GPT}, | ICSA 25 | Busca na página do ICSA 25 | **RQ1:** How accurate is GPT to answer questions about the architectural knowledge of an open-source system that is embedded in its model?<br>**RQ2:** How do software engineers evaluate the quality and trustworthiness of GPT's responses regarding the architectural knowledge of an open-source system that is embedded in its model? | | - Talvez fuja um pouco da questão de decisão de arquitetura por LLM.<br>- Faz um estudo exploratório com 14 Software Engineers<br>- Creio que possa ter outros estudos que estejam mais relacionados |
| Can LLMs Generate Architectural Design Decisions? - An Exploratory Empirical study | Architectural Knowledge Management (AKM) involves the organized handling of information related to architectural decisions and design within a project or organization. An essential artifact of AKM is the Architecture Decision Records (ADR), which documents key design decisions. ADRs are documents that capture decision context, decision made and various aspects related to a design decision, thereby promoting transparency, collaboration, and understanding. Despite their benefits, ADR adoption in software development has been slow due to challenges like time constraints and inconsistent uptake. Recent advancements in Large Language Models (LLMs) may help bridge this adoption gap by facilitating ADR generation. However, the effectiveness of LLM for ADR generation or understanding is something that has not been explored. To this end, in this work, we perform an exploratory study that aims to investigate the feasibility of using LLM for the generation of ADRs given the decision context. In our exploratory study, we utilize GPT and T5-based models with 0-shot, few-shot, and fine-tuning approaches to generate the Decision of an ADR given its Context. Our results indicate that in a 0-shot setting, state-of-the-art models such as GPT-4 generate relevant and accurate Design Decisions, although they fall short of human-level performance. Additionally, we observe that more cost-effective models like GPT-3.5 can achieve similar outcomes in a few-shot setting, and smaller models such as Flan-T5 can yield comparable results after fine-tuning. To conclude, this exploratory study suggests that LLM can generate Design Decisions, but further research is required to attain human-level generation and establish standardized widespread adoption. | [2403.01709] Can LLMs Generate Architectural Design Decisions? -An Exploratory Empirical study | R. Dhar, K. Vaidhyanathan and V. Varma, "Can LLMs Generate Architectural Design Decisions? - An Exploratory Empirical Study," in 2024 IEEE 21st International Conference on Software Architecture (ICSA), Hyderabad, India, 2024, pp. 79-89, doi: 10.1109/ICSA59870.2024.00016. keywords: {Software architecture;Large language models;Standards organizations;Collaboration; Organizations;Computer architecture; Knowledge management} url: https://doi.ieeecomputersociety.org/10.1109/ICSA59870.2024.00016 | ICSA 24 | Busca na página do ICSA 24 | **RQ1:** Can LLMs be successfully employed to generate architectural design decisions from a given context in a zero-shot setting?<br>**RQ2:** Does few-shot approach affect or improve a LLM's ability to generate Design Decisions?<br>**RQ3:** Does Fine-tuning LLMs enhances it's capability of generating architectural Design Decisions based on a provided context? | | - Creio que seja um trabalho bem relacionado basedo nas RQs e no abstract. |
| Rule-Based Architectural Design Pattern Recognition with GPT Models | Architectural design patterns are essential in software development because they offer proven solutions to large-scale structural problems in software systems and enable developers to create software that is more maintainable, scalable, and comprehensible. Model-View-Whatever (MVW) design patterns are prevalent in many areas of software development, but their use in Web development is on the rise. There are numerous subtypes of MVW design patterns applicable to Web systems, but there is no exhaustive listing of them. Additionally, it is unclear how these subtypes can be utilized in contemporary Web development, as their usage is typically unconscious. Here, we discuss and define the most prevalent MVW design patterns used in Web development, as well as provide Angular framework examples and guidance on when to employ a particular design pattern. On the premise of the primary characteristics of design patterns, we created a rule system that large language models (LLMs) can comprehend without doubt. Here, we demonstrate how effectively Generative Pre-trained Transformer (GPT) models can identify various design patterns based on our principles and verify the quality of our recommendations. Together, our solution and GPT models constitute an effective natural language processing (NLP) solution capable of detecting MVW design patterns in Angular projects with an average accuracy of 90%. | Rule-Based Architectural Design Pattern Recognition with GPT Models | Jánki, Z.R.; Bilicki, V. Rule-Based Architectural Design Pattern Recognition with GPT Models. Electronics 2023, 12, 3364. https://doi.org/10.3390/electronics12153364 | Electronics | Busca no google acadêmico por palavras chaves | **RQ1:** How precisely can MVW design pattern variations in Angular applications be detected?<br>**RQ2:** Which variation of the MVW design pattern is most prevalent in Angular application development?<br>**RQ3:** What design recommendations and best practices can be formulated for Angular applications?<br>**RQ4:** What additional opportunities do the results present? | | - Baseado nas RQs talvez o trabalho fuja um pouco do que eu esperava (vou dar uma lida por cima para confirmar), mas creio que tenham trabalhos mais relacionados. |
| An LLM-assisted approach to designing software architectures using ADD | Designing effective software architectures is a complex, iterative process that traditionally relies on expert judgment. This paper proposes an approach for Large Language Model (LLM)-assisted software architecture design using the Attribute-Driven Design (ADD) method. By providing an LLM with an explicit description of ADD, an architect persona, and a structured iteration plan, our method guides the LLM to collaboratively produce architecture artifacts with a human architect. We validate the approach through case studies, comparing generated designs against proven solutions and evaluating them with professional architects. Results show that our LLM-assisted ADD process can generate architectures closely aligned with established solutions and partially satisfying architectural drivers, highlighting both the promise and current limitations of using LLMs in architecture design. Our findings emphasize the importance of human oversight and iterative refinement when leveraging LLMs in this domain. | An LLM-assisted approach to designing software architectures using ADD | X | Aparentemente sem publicação | Busca no google acadêmico por palavras chaves | **RQ1:** Does an LLM-assisted ADD design result in a solution that is similar to a proven solution?<br>**RQ2:** Does an LLM-assisted ADD design produce architectures that satisfy the architectural drivers?<br>**RQ3:** Does the use of ADD make a significant difference when asking an LLM to design an architecture? | | - Trabalho aparentemente bem relacionado, com etapas similares as que estamos propondo<br>- Desvantagem: aparentemente o trabalho não foi publicado. |
| Using LLMs in Generating Design Rationale for Software Architecture Decisions | Design Rationale (DR) for software architecture decisions refers to the reasoning underlying architectural choices, which provides valuable insights into the different phases of the architecting process throughout software development. However, in practice, DR is often inadequately documented due to a lack of motivation and effort from developers. With the recent advancements in Large Language Models (LLMs), their capabilities in text comprehension, reasoning, and generation may enable the generation and recovery of DR for architecture decisions. In this study, we evaluated the performance of LLMs in generating DR for architecture decisions. First, we collected 50 Stack Overflow (SO) posts, 25 GitHub issues, and 25 GitHub discussions related to architecture decisions to construct a dataset of 100 architecture-related problems. Then, we selected five LLMs to generate DR for the architecture decisions with three prompting strategies, including zero-shot, chain of thought (CoT), and LLM-based agents. With the DR provided by human experts as ground truth, the Precision of LLM-generated DR with the three prompting strategies ranges from 0.267 to 0.278, Recall from 0.627 to 0.715, and F1-score from 0.351 to 0.389. Additionally, 64.45% to 69.42% of the arguments of DR not mentioned by human experts are also helpful, 4.12% to 4.87% of the arguments have uncertain correctness, and 1.59% to 3.24% of the arguments are potentially misleading. To further understand the trustworthiness and applicability of LLM-generated DR in practice, we conducted semi-structured interviews with six practitioners. Based on the experimental and interview results, we discussed the pros and cons of the three prompting strategies, the strengths and limitations of LLM-generated DR, and the implications for the practical use of LLM-generated DR. | Using LLMs in Generating Design Rationale for Software Architecture Decisions | ACM Transactions on Software Engineering and Methodology | Zhou, Xiyu, et al. "Using LLMs in generating design rationale for software architecture decisions." ACM Transactions on Software Engineering and Methodology 35.8 (2026): 1-38. | ACM Transactions on Software Engineering and Methodology | Busca no google acadêmico por palavras chaves | **RQ1:** How effective are LLMs in generating and recovering design rationale for architecture decisions?<br>**RQ2:** Does the CoT technique affect or enhance the capability of LLMs to generate and recover design rationale for architecture decisions?<br>**RQ3:** Do LLM-based agents affect or enhance the capability of LLMs to generate and recover design rationale for architecture decisions? | | - Aparentemente tem outros focos, mais voltado para técnicas de prompt, mas creio que  possam se relacionar |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Better Together: An LLM-assisted approach to designing software architectures using Attribute-Driven Design | Designing effective software architectures is a complex, iterative process that traditionally relies on expert judgment. This paper proposes an approach for Large Language Model (LLM)-assisted software architecture design leveraging the Attribute-Driven Design (ADD) method. By providing an LLM with an explicit description of ADD and an architect persona, our method guides the LLM to collaboratively produce a structured iteration plan and a detailed architecture design document. We have validated this approach in two ways: through the comparison of an LLM-generated solution with a proven solution, and through the evaluation of an LLM-generated architecture by a team of four professional architects. We also compare designs produced with and without our ADD-guided approach. Our results show that our LLM-assisted ADD process produces thorough architecture documentation and that the LLM-generated architectures satisfy architectural drivers. We also argue that human feedback and guidance are still needed in the design process, particularly with respect to the most critical and nuanced design decisions. | Better Together: An LLM-assisted approach to designing software architectures using Attribute-Driven Design \| IEEE Journals & Magazine \| IEEE Xplore | | | Busca no google acadêmico por palavras chaves | | Não consegui baixar | |